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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4442 Service and Cloud Computing

Session : 2024 / 2025 Semester II

Date : 13 May 2025 Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Examiner(s): Dr ZHANG Zhaorui

This question paper has a total of ___6___ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- This is an open-book exam.
- The exam is 100 marks in total.
- Any student caught cheating, e.g., sharing or copying answers with classmates, will receive a ZERO mark for the exam.
- Write your answers in the **Answer Book** and submit it **together with the question paper**.

Student Name: _____ Student No.: _____

Do not turn over the page until you are told to do so!

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QUESTION 2 (15 MARKS)

Short Questions:

(1) What is the primary difference between Infrastructure as a Service (IaaS) and Platform as a Service (PaaS) in cloud computing? (3 marks)

(2) What is the purpose of a hypervisor in cloud computing? (3 marks)

(3) Please introduce the main components of the Kubernetes and their roles and functions. (3 marks)

(4) Please describe the relationship between Hadoop and Spark. (3 marks)

(5) What is the difference between a "transformation" and an "action" in Spark? (3 marks)

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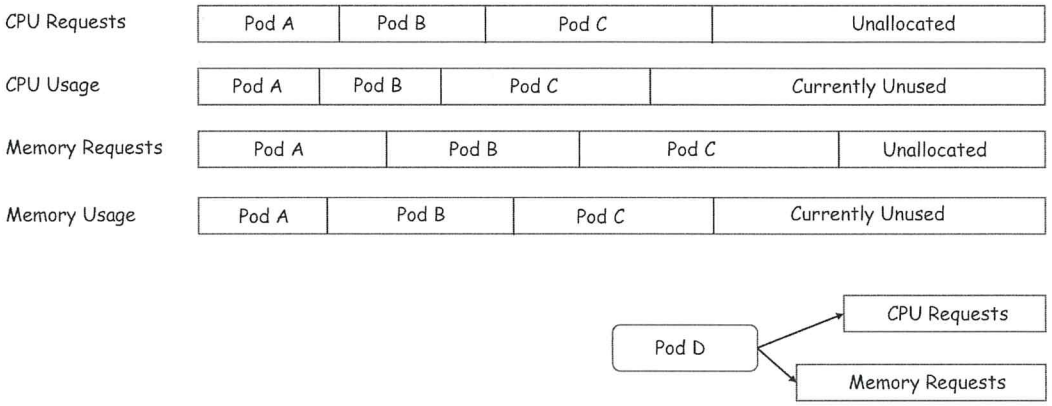
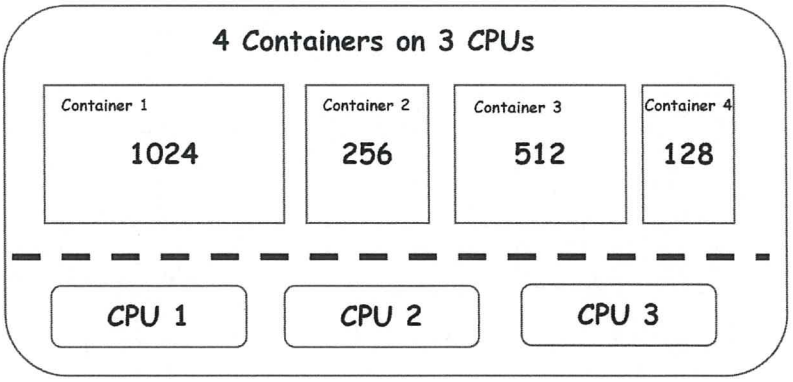
QUESTION 3 (15 MARKS)

Docker is a platform for developing, shipping, and running applications in containers. Containers package an application with its dependencies, ensuring consistency across environments. Docker simplifies deployment, enhances scalability, and improves resource efficiency. Kubernetes is an open-source system for automating the deployment, scaling, and management of containerized applications. It organizes containers into pods and provides features like load balancing and self-healing, enabling efficient and resilient application management at scale. Together, Docker and Kubernetes form a powerful combination for modern cloud-native application development, enabling developers to build applications that are scalable, portable, and easy to manage across diverse cloud infrastructures. Please answer the following questions for docker and Kubernetes:



a) For the above figure, we assume the bar is available CPU and Memory for a certain computing node. We want to deploy several Pods of K8s on this computing node. The CPU requests mean the total CPUs for each pod request. The CPU usage is the total CPUs that have been used currently for each pod. The Memory requests and usage are same with CPUs. Please answer that if we want to deploy the Pod D on this computing node, can we deploy the Pod D on this computing node, if this computing node has enough resources to deploy the Pod D, and also please give a detailed explanation about why we can/can't deploy it on this computing node. (8 marks)

b) According to the following figure, we want to deploy 4 docker containers on a certain computing node that has 3 CPUs. Please provide a calculation formula and calculate how many CPUs each container can use. (7 marks)



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QUESTION 4 (20 MARKS)

Apache Hadoop is an open-source framework designed for distributed storage and processing of large datasets across clusters of computers. It uses a simple programming model and is designed to scale from a single server to thousands of machines. Hadoop consists of three main components:

1. Hadoop Distributed File System (HDFS)
2. MapReduce.
3. Yarn.

Hadoop is widely used for big data analytics, enabling organizations to store and analyze vast amounts of data efficiently and cost-effectively. Please answer the following questions:

a) Please provide the workflow for how Hadoop works for a certain MapReduce (based on Hadoop) application. (8 marks)

b) Please provide an explanation for the three main components of Hadoop, including HDFS, MapReduce, and Yarn. (6 marks)

c) Please provide a description of "Application Master," "Resource Manager," "Node Manager," "NameNode," and "DataNode." Please provide a detailed description of them and also their roles and functions in the Hadoop application. (6 marks)

QUESTION 5 (20 MARKS)

Kubernetes, also known as K8S, is an open-source platform for managing containerized workloads and services. It was originally designed by Google and is now maintained by the Cloud Native Computing Foundation.

a) Previously, Docker is often used to create containers for Kubernetes. For the Docker, please describe what the "Cgroups" and "NameSpaces" are. (5 marks)

b) Explain the architecture of Kubernetes. (7 marks)

c) Describe the main workflow of the Kubernetes and also draw a figure to show them. (8 marks)

QUESTION 6 (20 MARKS)

A Spark Python program based on RDDs is a program that uses the PySpark API to perform distributed computing tasks on RDDs in Python. The program typically follows a series of steps, including loading data from a data source, transforming the data using RDD operations like map, filter, and reduceByKey, and finally, writing the results to an output storage system. Spark Python programs based on RDDs are typically used for processing and analyzing large-scale data sets in a distributed and fault-tolerant manner.

We provide a Spark program in Python using PySpark to analyze a dataset of e-commerce transactions stored in a JSON file. The dataset contains fields: transaction_id, user_id, product_id, category, amount, and timestamp. please answer the following questions.

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Spark Code:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, sum as _sum, avg

spark = SparkSession.builder.appName("ECommerceAnalysis").getOrCreate()

df = spark.read.json("path/to/transactions.json")

revenue_per_category = (df
                .groupBy("category")
                .agg(_sum("amount").alias("total_revenue")))

top_3_categories = (revenue_per_category
            .orderBy(col("total_revenue").desc())
            .limit(3))

top_category_transactions = df.join(top_3_categories, "category")
average_transaction_amount = (top_category_transactions
                    .groupBy("category")
                    .agg(avg("amount").alias("average_transaction_amount")))

top_3_categories.show()
average_transaction_amount.show()

spark.stop()
```

Please answer the following questions based on the above description, dataset, and Spark Code:

a) Please provide a description for each line of the code based on the knowledge of Spark and RDD. (5 marks)
b) What is the "Transformation" in Spark? Describe the "Transformation" and provide the example from the above Spark code. (5 marks)
c) What is the "Action" in Spark? Describe the "Action" and provide the example from the above Spark code. (5 marks)
d) What are the main components of the Spark, and explain each component? (5 marks)

END